

Information Theory for ML: Entropy, KL Divergence, and Why Cross-Entropy Loss Is Maximum Likelihood

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The quantities behind every “cross-entropy loss” line of ML code, and the identity that ties this whole repository together: minimizing NLL is minimizing a KL divergence. `information.theory.py` verifies each claim numerically.

1 Entropy

Definition 1. For a distribution p on a finite alphabet,

$$H(p) = - \sum_x p(x) \log p(x) \quad (0 \log 0 := 0). \quad (1)$$

H measures the uncertainty of a draw — equivalently (source coding theorem) the optimal average code length, in nats for $\log_e$, bits for $\log_2$. Extremes: $H = 0$ iff p is deterministic; on k symbols H is maximized by the uniform distribution at $\log k$. The Bernoulli curve $H(p) = -p \log p - (1-p) \log(1-p)$ peaks at $p = \frac{1}{2}$ (script, left panel). Among densities on $\mathbb{R}$ with fixed variance, the Gaussian maximizes (differential) entropy — one reason it is the default noise model.

2 KL divergence

Definition 2. For $p \ll q$ (i.e. $q(x) = 0 \Rightarrow p(x) = 0$),

$$\text{KL}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)}. \quad (2)$$

The expected excess code length for encoding p -data with a code built for q — the natural “how wrong is model q about reality p .”

Theorem 1 (Gibbs’ inequality). $\text{KL}(p \parallel q) \geq 0$, with equality iff $p = q$.

Proof. With $\log t \leq t - 1$ (equality iff $t = 1$), for x with $p(x) > 0$: $-\text{KL} = \sum_x p(x) \log \frac{q(x)}{p(x)} \leq \sum_x p(x) \left(\frac{q(x)}{p(x)} - 1 \right) = \sum_x q(x) - 1 \leq 0$, with equality iff $q(x) = p(x)$ everywhere p charges. $\square$

Not a metric: asymmetric, no triangle inequality. The direction matters — $\text{KL}(p \parallel q)$ punishes q for assigning tiny mass where p lives (mode-covering); $\text{KL}(q \parallel p)$ tolerates dropping modes.

3 Cross-entropy, and the identity behind the loss

Definition 3. $H(p, q) = -\sum_x p(x) \log q(x)$.

Adding and subtracting $H(p)$ gives the decomposition

$$\boxed{H(p, q) = H(p) + \text{KL}(p \parallel q).} \quad (3)$$

Since $H(p)$ does not involve q : *minimizing cross-entropy over the model q is minimizing the KL divergence to it*, and $H(p)$ is the irreducible floor of the loss.

4 Maximum likelihood is KL minimization

Let $\hat{p}$ be the *empirical distribution* of a sample $x_1, \dots, x_n$ (mass $\frac{1}{n}$ at each observation). For any model q_θ , the average NLL rewrites as

$$\frac{1}{n} \sum_{i=1}^n -\log q_\theta(x_i) = -\sum_x \hat{p}(x) \log q_\theta(x) = H(\hat{p}, q_\theta) \stackrel{(3)}{=} H(\hat{p}) + \text{KL}(\hat{p} \parallel q_\theta). \quad (4)$$

The θ -dependence sits entirely in the KL term, hence

$$\boxed{\arg \min_{\theta} \text{NLL}(\theta) = \arg \min_{\theta} \text{KL}(\hat{p} \parallel q_\theta) :} \quad (5)$$

maximum likelihood fits the model to the data's empirical distribution in the KL sense. Every MLE in this repository — Bernoulli, Gaussian, least squares — is an instance; the script overlays $\text{NLL}(q)$ and $H(\hat{p}) + \text{KL}(\hat{p} \parallel q)$ for Bernoulli and the curves coincide to 10^{-12} , with the minimum at $\bar{x}$ sitting on the entropy floor. (This also explains why the loss never reaches zero: the floor is $H(\hat{p})$, the data's own uncertainty.)

5 Mutual information

Definition 4. $I(X; Y) = \text{KL}(p_{XY} \parallel p_X p_Y) = H(X) - H(X \mid Y) = H(X) + H(Y) - H(X, Y)$.

The KL distance of the joint from independence — how many nats one variable carries about the other; zero iff independent, and symmetric despite KL's asymmetry. It upper-bounds what any predictor of Y from X can achieve (Fano), and appears across ML as the objective of representation learning (InfoNCE, information bottleneck) and the capacity limit of channels. Worth internalizing next to (5): entropy is the floor of prediction, mutual information is the budget of relevance.